

“PREDICTIVE ANALYTICS FOR BREAST CANCER DIAGNOSIS USING MACHINE LEARNING”

*A Project submitted to the University of Kerala in partial fulfilment of the
requirements for the award of Master of Science in Statistics*



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CERTIFICATE

This is to certify that the report entitled “**Predictive Analytics for Breast Cancer Diagnosis Using Machine Learning**” is a bonafide record of work done by ASEEM S, Reg No. 62524144007, M.Sc. Statistics student in the University College, Thiruvananthapuram, during 2024-2026. He has carried out the work under my supervision and guidance in fulfilment of the dissertation.

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DECLARATION

I hereby declare that the report entitled “**Predictive Analytics for Breast Cancer Diagnosis Using Machine Learning**” submitted in partial fulfilment of the project requirement for the degree of M.Sc. Statistics, University College, Thiruvananthapuram, is a record of original work carried out by me under the guidance of Mrs Neena Nelson, Guest Lecturer, Department of Statistics, University College, Thiruvananthapuram. The work has not previously formed the basis for the award of any degree, diploma, fellowship, or other title or recognition.

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ABSTRACT

Breast cancer is one of the most common cancers affecting women worldwide and remains a major public health concern. Early diagnosis and timely treatment are essential for improving survival rates and reducing mortality. In recent years, machine learning techniques have been increasingly applied in healthcare to support disease prediction and diagnosis.

The objective of this study is to develop a machine learning model for breast cancer prediction using the Wisconsin Breast Cancer Dataset. The dataset consists of 569 observations and 32 attributes, including the diagnosis variable and diagnostic features extracted from digitized images of breast cell nuclei. Data preprocessing techniques were applied, followed by train-test splitting using an 80:20 ratio. A supervised machine learning classification model was implemented using Python. The Wisconsin Breast Cancer Dataset was downloaded from Kaggle and uploaded to Google Colab for preprocessing, model training, and evaluation.

The performance of the model was evaluated using accuracy, confusion matrix, precision, recall, and F1-score. The developed model achieved an accuracy of **95.6%**, indicating excellent classification performance in distinguishing between benign and malignant breast tumors. Various visualizations, including count plots, pie charts, correlation heatmaps, confusion matrices, and feature importance plots, were used to analyze the dataset and interpret model performance.

The findings demonstrate that machine learning techniques can effectively support breast cancer diagnosis and assist healthcare professionals in clinical decision-making. The study highlights the potential of predictive analytics as a reliable and cost-effective tool for early cancer detection.

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CHAPTER 1

INTRODUCTION

1.1 BREAST CANCER OVERVIEW

Breast cancer is one of the most common cancers affecting women worldwide. It occurs when abnormal cells in the breast grow uncontrollably and form a tumor. According to global health reports, breast cancer is a major cause of cancer-related deaths among women. However, advancements in medical technology and early detection methods have significantly improved survival rates. Early diagnosis and appropriate treatment can greatly increase the chances of recovery.

1.2 WHAT IS BREAST CANCER?

Breast cancer is a disease in which cells in the breast tissue divide and grow abnormally. These cancerous cells can invade surrounding tissues and may spread to other parts of the body through the blood or lymphatic system, a process known as metastasis. Breast cancer can occur in both men and women, although it is much more common in women.

1.3 TYPES OF BREAST CANCER

Breast cancer can be classified into different types based on the location and behavior of cancer cells. The major types include:

1.3.1 DUCTAL CARCINOMA IN SITU (DCIS)

A non-invasive cancer where abnormal cells are found in the lining of the milk ducts.

1.3.2 INVASIVE DUCTAL CARCINOMA (IDC)

The most common type of breast cancer, which begins in the milk ducts and spreads to nearby breast tissues.

1.3.3 INVASIVE LOBULAR CARCINOMA (ILC)

This cancer begins in the milk-producing lobules and can spread to surrounding tissues.

1.3.4 TRIPLE-NEGATIVE BREAST CANCER

An aggressive form of breast cancer that lacks estrogen, progesterone, and HER2 receptors.

1.3.5 INFLAMMATORY BREAST CANCER

A rare but fast-growing type of breast cancer that causes redness and swelling of the breast.

1.4 RISK FACTORS OF BREAST CANCER

Several factors increase the likelihood of developing breast cancer. These include:

- Increasing age
- Family history of breast cancer
- Genetic mutations such as BRCA1 and BRCA2
- Hormonal changes
- Obesity and overweight
- Lack of physical activity
- Smoking and alcohol consumption
- Exposure to radiation
- Late menopause or early menstruation

Although these factors may increase risk, breast cancer can also occur in individuals without any known risk factors.

1.5 IMPORTANCE OF EARLY DETECTION

Early detection plays a crucial role in reducing breast cancer mortality. When breast cancer is identified at an early stage, treatment options are more effective and less invasive. Early diagnosis improves survival rates, reduces treatment costs, and enhances the quality of life for patients.

Traditional diagnostic methods such as mammography, ultrasound, biopsy, and clinical examinations are commonly used for breast cancer detection. However, these methods may require significant time and expert interpretation. Therefore, advanced computational

techniques are increasingly being used to assist healthcare professionals in making accurate and timely diagnoses.

1.6 MACHINE LEARNING IN HEALTHCARE

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make predictions without being explicitly programmed. In healthcare, machine learning is widely used for disease diagnosis, medical image analysis, patient risk assessment, drug discovery, and treatment planning.

Machine learning techniques can analyze large volumes of medical data and identify hidden patterns that may not be easily detected by humans. As a result, ML has become an important tool for supporting clinical decision-making.

1.7 SUPERVISED LEARNING

Supervised learning is one of the most commonly used machine learning approaches. In supervised learning, a model is trained using labeled data, where the correct output is already known. The model learns the relationship between input features and output labels and then predicts outcomes for new data.

In breast cancer prediction, supervised learning algorithms are trained using datasets containing medical measurements and corresponding diagnoses (benign or malignant). The trained model can then classify new cases based on learned patterns.

1.8 CLASSIFICATION PROBLEMS

Classification is a machine learning task that involves assigning data into predefined categories. In the context of breast cancer prediction, classification algorithms determine whether a tumor is:

- Benign (non-cancerous)
- Malignant (cancerous)

Common classification algorithms include:

- Logistic Regression

- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Naïve Bayes

These algorithms are evaluated based on performance metrics such as accuracy, precision, recall, and F1-score.

1.9 HEALTHCARE APPLICATIONS OF MACHINE LEARNING

Machine learning has numerous applications in healthcare, including:

- Disease diagnosis and prediction
- Medical image processing
- Cancer detection
- Patient monitoring systems
- Drug development
- Personalized treatment recommendations
- Hospital resource management

The integration of machine learning into healthcare systems helps improve diagnostic accuracy and supports evidence-based medical decisions.

1.10 NEED FOR THE STUDY

Breast cancer remains a major public health concern worldwide. Accurate and timely diagnosis is essential for improving patient outcomes and reducing mortality rates. Traditional diagnostic approaches may sometimes be time-consuming and dependent on expert interpretation.

Machine learning offers an efficient and reliable approach for analyzing medical data and predicting breast cancer outcomes. By developing predictive models using machine learning algorithms, healthcare professionals can receive additional support in identifying cancer cases at an early stage.

Therefore, this study aims to explore the application of machine learning techniques for breast cancer prediction and evaluate their effectiveness in improving diagnostic accuracy and supporting early detection.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

Breast cancer is one of the leading causes of cancer-related deaths among women worldwide. Early detection and accurate diagnosis play a vital role in improving patient survival rates. In recent years, machine learning techniques have gained significant attention in the healthcare sector due to their ability to analyze large amounts of medical data and provide accurate predictions. Numerous researchers have applied machine learning algorithms to breast cancer datasets to develop predictive models capable of distinguishing between benign and malignant tumors.

This chapter reviews previous studies related to breast cancer prediction using machine learning algorithms, with particular emphasis on the Wisconsin Breast Cancer Dataset.

2.2 BREAST CANCER DIAGNOSIS USING MACHINE LEARNING

Machine learning has become an important tool for medical diagnosis because of its ability to identify complex patterns in clinical data. Researchers have applied various supervised learning algorithms to classify breast cancer cases and improve diagnostic accuracy.

Several studies have demonstrated that machine learning models can achieve high classification performance when trained using well-structured medical datasets. These models assist healthcare professionals in making faster and more reliable decisions during diagnosis.

2.3 WISCONSIN BREAST CANCER DATASET

The Wisconsin Breast Cancer Dataset (WBCD) is one of the most widely used datasets for breast cancer classification studies. The dataset was developed by researchers at the University of Wisconsin and is publicly available through the UCI Machine Learning Repository and the Scikit-learn library.

The dataset contains features extracted from digitized images of fine needle aspirate (FNA) samples of breast masses. These features describe various characteristics of cell nuclei, including:

- Radius
- Texture
- Perimeter
- Area
- Smoothness
- Compactness
- Concavity
- Symmetry
- Fractal Dimension

The target variable classifies tumors into two categories:

- Benign (B)
- Malignant (M)

Due to its reliability and balanced structure, the Wisconsin Breast Cancer Dataset has been extensively used for evaluating machine learning classification algorithms.

2.4 PREVIOUS RESEARCH STUDIES

STUDY 1: LOGISTIC REGRESSION FOR BREAST CANCER PREDICTION

Researchers have employed Logistic Regression as a baseline classification technique for breast cancer diagnosis. The algorithm estimates the probability of a tumor being malignant based on predictor variables. Studies have reported high classification accuracy due to the dataset's well-defined feature structure. Logistic Regression is valued for its simplicity, interpretability, and computational efficiency.

STUDY 2: DECISION TREE-BASED CLASSIFICATION

Decision Tree algorithms have been widely used for breast cancer classification because they provide clear decision rules and easy interpretation. Researchers found that Decision Trees can effectively classify tumors by selecting the most informative features at each decision node. However, performance may decrease when the model becomes overly complex.

STUDY 3: RANDOM FOREST APPROACH

Random Forest is an ensemble learning method that combines multiple Decision Trees to improve prediction accuracy. Several studies have shown that Random Forest achieves excellent performance on the Wisconsin Breast Cancer Dataset by reducing overfitting and improving generalization capability. The algorithm is often considered one of the most reliable classifiers for medical diagnosis.

STUDY 4: SUPPORT VECTOR MACHINE (SVM)

Support Vector Machine is among the most popular machine learning algorithms used in cancer prediction. Researchers have demonstrated that SVM can effectively separate benign and malignant cases by identifying an optimal decision boundary. Many studies have reported classification accuracies above 95% when using SVM on the Wisconsin Breast Cancer Dataset.

STUDY 5: K-NEAREST NEIGHBORS (KNN)

K-Nearest Neighbors is a distance-based classification algorithm that assigns class labels based on neighboring observations. Studies indicate that KNN performs well on breast cancer datasets when appropriate feature scaling techniques are applied. The algorithm is simple yet effective for binary classification problems.

2.5 COMPARATIVE ANALYSIS OF MACHINE LEARNING ALGORITHMS

Various machine learning algorithms have been compared by researchers to identify the most effective technique for breast cancer diagnosis. The comparison is generally based on performance measures such as:

- Accuracy
- Precision

- Recall
- F1-Score
- Confusion Matrix

Research findings indicate that ensemble methods and Support Vector Machines often provide superior classification performance. However, simpler models such as Logistic Regression also achieve competitive results while offering better interpretability.

2.6 PYTHON FOR MACHINE LEARNING APPLICATIONS

Python has become one of the most widely used programming languages for machine learning applications due to its simplicity and extensive library support. Libraries commonly used in breast cancer prediction studies include:

- NumPy
- Pandas
- Matplotlib
- Scikit-learn

Scikit-learn provides built-in access to the Wisconsin Breast Cancer Dataset and offers efficient implementations of various classification algorithms. Researchers frequently utilize Python to preprocess data, train models, evaluate performance, and visualize results.

2.7 RESEARCH GAP

Although many studies have achieved high classification accuracy using machine learning techniques, there is still a need for developing models that are accurate, computationally efficient, and easy to interpret. Different algorithms may perform differently depending on dataset characteristics and preprocessing methods.

Therefore, further evaluation of machine learning algorithms on the Wisconsin Breast Cancer Dataset is necessary to identify suitable models for supporting early breast cancer diagnosis.

2.8 SUMMARY

The literature review demonstrates that machine learning techniques have significantly improved breast cancer diagnosis and prediction. The Wisconsin Breast Cancer Dataset has served as a standard benchmark dataset for evaluating classification algorithms. Previous studies have reported high accuracy using algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbors.

In the present study, machine learning techniques are applied to the Wisconsin Breast Cancer Dataset using Python. The developed model achieved an accuracy of approximately 95.6%, indicating its effectiveness in classifying breast tumors as benign or malignant. The findings support the growing role of machine learning in assisting healthcare professionals with early breast cancer detection and diagnosis.

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

This chapter describes the methodology adopted for developing a machine learning model for breast cancer prediction. The study utilizes the Wisconsin Breast Cancer Dataset and applies machine learning techniques to classify tumors as benign or malignant. The methodology includes data collection, data preprocessing, model development, training, testing, and performance evaluation.

3.2 RESEARCH METHODOLOGY

The overall workflow of the study consists of the following steps:

1. Data Collection
2. Data Preprocessing
3. Exploratory Data Analysis
4. Feature Selection
5. Data Splitting
6. Model Training
7. Model Testing
8. Performance Evaluation

The machine learning model is developed using Python and relevant data science libraries.

FIGURE 3.1: METHODOLOGY FLOWCHART

Figure 3.1 illustrates the overall workflow adopted in the present study. The process begins with dataset collection, followed by data preprocessing, exploratory data analysis, feature selection, train-test splitting, model training, prediction, and performance evaluation. This systematic approach ensures accurate and reliable breast cancer classification.

The breast cancer diagnosis model was developed following a structured machine learning pipeline, which can be broken down into several key stages:

1. Data Loading and Initial Preprocessing : The process began by loading the breast cancer dataset into a Pandas DataFrame. Initial preprocessing involved removing irrelevant columns such as 'id' and 'Unnamed: 32' (if empty), to ensure the data was clean and ready for analysis.

2. Exploratory Data Analysis (EDA) : Comprehensive EDA was performed to understand the dataset's characteristics. This included:

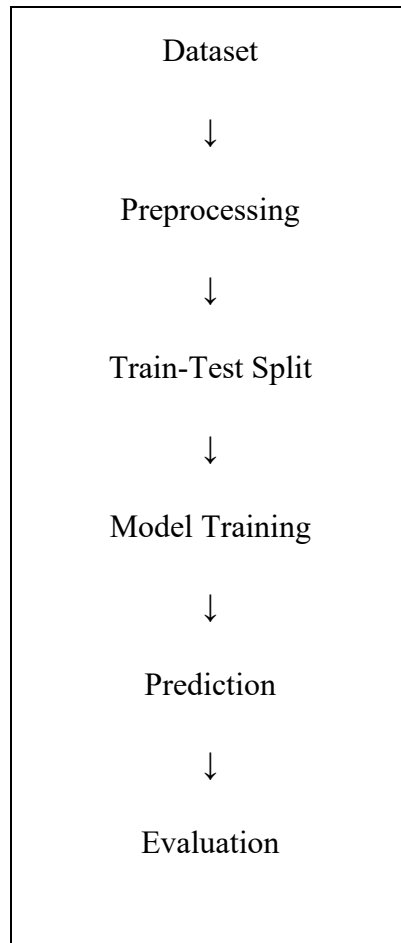
- Histograms : Visualizing the distribution of individual numerical features to identify their spread and potential skewness.
- Correlation Heatmap : Analyzing the relationships and correlations between different numerical features to detect multicollinearity and understand feature dependencies.
- Benign vs Malignant Count Plot : A bar chart to show the absolute counts of benign (B) and malignant (M) diagnoses, revealing class imbalance if any.
- Diagnosis Distribution Pie Chart : A pie chart illustrating the proportion of benign and malignant cases, providing a clear visual representation of the class distribution.

3. Feature Engineering and Data Preparation : The 'diagnosis' column, representing the target variable, was label encoded (M=1, B=0) to convert categorical labels into numerical format suitable for machine learning algorithms. The dataset was then split into features (X) and the target (y). This was followed by splitting the data into training and testing sets to prepare for model training and unbiased evaluation.

4. Model Training : A RandomForestClassifier model was initialized and trained using the prepared training data (X_train, y_train). This ensemble method is robust and generally performs well on classification tasks.

5. Model Evaluation : After training, the model's performance was rigorously evaluated using the test set (X_{test} , y_{test}). Key evaluation metrics and visualizations included:

- Accuracy Score : A quantitative measure of the model's overall correctness.
- Confusion Matrix : A heatmap visualizing the true positives, true negatives, false positives, and false negatives, providing detailed insights into the model's classification performance for each class.
- Feature Importance Plot : A bar chart displaying the importance of each feature in the RandomForestClassifier's decision-making process, helping to identify the most influential factors for breast cancer diagnosis.



3.3 DATASET DESCRIPTION

The dataset used in this study was downloaded from Kaggle and uploaded to Google Colab for analysis. The dataset contains 569 observations and 32 attributes, including an identification number, diagnosis label, and diagnostic features computed from digitized images of breast cell nuclei. The diagnosis variable classifies tumors as Benign (B) or Malignant (M).

DATASET CHARACTERISTICS

- Total Number of Instances: 569
- Number of Features: 30
- Number of Classes: 2
- Target Classes:
 - Malignant (Cancerous)
 - Benign (Non-Cancerous)

The dataset contains numerical features computed from digitized images of breast mass cell nuclei.

Dataset Characteristics Table

Characteristic	Value
Dataset Name	Wisconsin Breast Cancer Dataset
Source	Kaggle
Total Records	569
Total Attributes	32
Target Variable	Diagnosis
Classes	Benign (B), Malignant (M)

Table 3.1 Dataset Characteristics

Sample Features

- Mean Radius
- Mean Texture
- Mean Perimeter

- Mean Area
- Mean Smoothness
- Mean Compactness
- Mean Concavity
- Mean Symmetry
- Fractal Dimension

These features are used as predictor variables for classification.

3.4 SOFTWARE AND TOOLS USED

The implementation was carried out using Python programming language.

Software

- Python 3.x
- Jupyter Notebook

Libraries Used

- NumPy
- Pandas
- Matplotlib
- Scikit-learn

These libraries provide functionalities for data handling, visualization, machine learning model development, and evaluation.

3.5 DATA COLLECTION

The dataset was downloaded from Kaggle, a widely used platform for data science and machine learning projects. The dataset contains breast cancer diagnostic measurements derived from digitized images of fine needle aspirates (FNA) of breast masses. The dataset includes 569 records and is suitable for binary classification tasks. After downloading the dataset in CSV format, it was uploaded to Google Colab for preprocessing, analysis, model training, and

evaluation. The data was then preprocessed and analyzed using Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

Import pandas as pd

```
df = pd.read_csv('breast-cancer-wisconsin-data.csv')
```

```
print(df.head())
```

```
print(df.shape)
```

The dataset consists of input features and target labels representing breast cancer diagnoses.

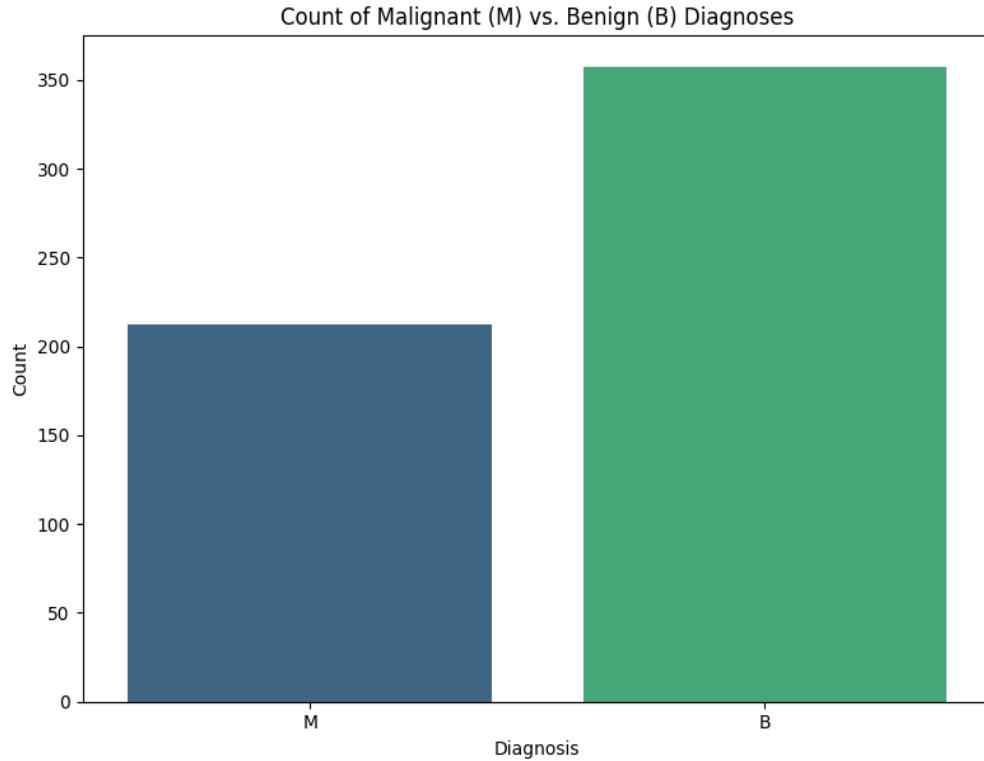
SOFTWARE AND TOOLS USED

Software/Tool	Purpose
Python	Programming Language
Google Colab	Development Environment
Pandas	Data Manipulation
NumPy	Numerical Computation
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Scikit-learn	Machine Learning Algorithms

Table 3.2 Software and Tools Used

FIGURE 3.2: BENIGN VS MALIGNANT COUNT PLOT

FIGURE 3.2: DISTRIBUTION OF BENIGN AND MALIGNANT TUMORS

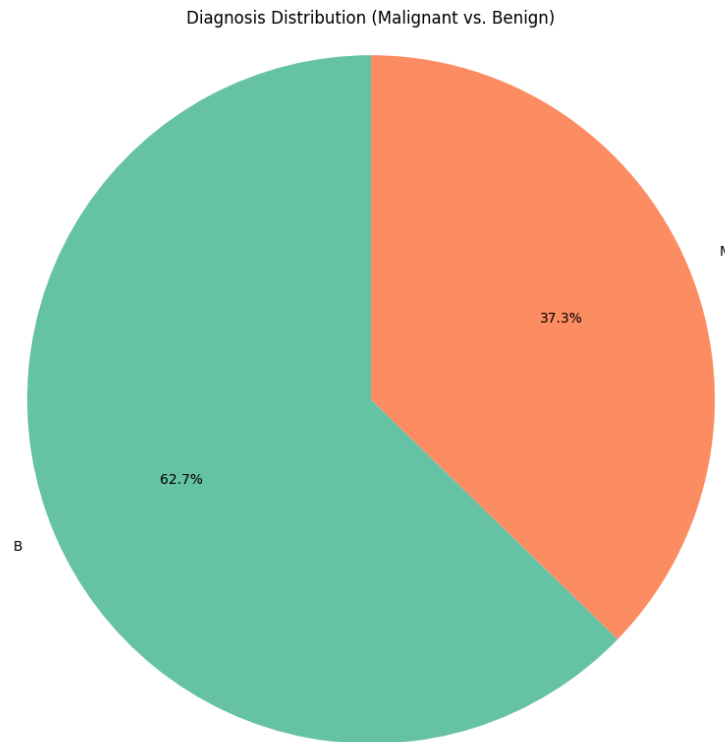


INTERPRETATION:

The count plot shows the distribution of diagnosis categories in the Wisconsin Breast Cancer Dataset. Out of 569 observations, 357 cases (62.7%) are benign and 212 cases (37.3%) are malignant. Although benign cases are more frequent, both classes are adequately represented, making the dataset suitable for machine learning classification.

FIGURE 3.3: DIAGNOSIS DISTRIBUTION OF BREAST CANCER CASES (PIE CHART)

FIGURE 3.3: PERCENTAGE DISTRIBUTION OF BENIGN AND MALIGNANT CASES



INTERPRETATION:

The pie chart presents the percentage distribution of diagnosis categories in the Wisconsin Breast Cancer Dataset. Benign tumors constitute approximately **62.7%** of the total observations, while malignant tumors account for **37.3%**. The chart indicates that benign cases are more prevalent within the dataset.

However, the presence of a substantial number of malignant cases ensures that the dataset remains suitable for machine learning classification. The visualization provides a clear understanding of the class composition and highlights the importance of accurately identifying malignant tumors for early diagnosis and treatment.

Key Insights

- Benign tumors represent the majority of cases in the dataset.
- Malignant tumors account for more than one-third of all observations.
- Both classes are sufficiently represented for classification analysis.
- The dataset does not exhibit severe class imbalance.
- Understanding class distribution is important before model training and evaluation.

3.6 DATA PREPROCESSING

Data preprocessing is an important step in machine learning because model performance depends heavily on data quality.

The following preprocessing procedures were performed:

3.6.1 LOADING DATASET

The dataset was imported into a Pandas DataFrame for analysis.

3.6.2 CHECKING MISSING VALUES

The dataset was examined for missing values.

`df.isnull().sum()`

No missing values were found in the dataset.

3.6.3 FEATURE AND TARGET SEPARATION

The predictor variables and target variable were separated.

```
X = df.drop('diagnosis', axis=1)
```

```
y = df['diagnosis']
```

Where:

- X = Input Features
- y = Target Variable

3.7 DATA SPLITTING

The dataset was divided into training and testing datasets.

Training Set

80% of the data was used for model training.

Testing Set

20% of the data was used for model evaluation.

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.20,  
    random_state=42  
)
```

The `random_state` parameter ensures reproducibility of results.

3.8 MACHINE LEARNING MODEL

A supervised machine learning classification algorithm was used to predict whether a tumor is benign or malignant.

The model learns patterns from the training dataset and predicts outcomes for unseen observations.

Model Training

```
model.fit(X_train, y_train)
```

The training process allows the algorithm to learn relationships between the input features and target classes.

Model Prediction

```
y_pred = model.predict(X_test)
```

Predictions were generated for the testing dataset.

3.9 PERFORMANCE EVALUATION

The model's performance was evaluated using classification metrics.

3.9.1 ACCURACY

Accuracy measures the proportion of correctly classified observations.

Formula:

$$\text{Accuracy} = (\text{Correct Predictions} / \text{Total Predictions}) \times 100$$

In Python:

```
from sklearn.metrics import accuracy_score
```

```
accuracy = accuracy_score(y_test, y_pred)
```

Obtained Accuracy

The developed model achieved an accuracy of:

95.6%

(Accuracy = 0.956140350877193)

This indicates that approximately 96 out of every 100 cases were classified correctly.

3.9.2 CONFUSION MATRIX

A confusion matrix was used to compare actual and predicted classifications.

It provides information about:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred)

3.9.3 CLASSIFICATION REPORT

A classification report was generated to evaluate:

- Precision
- Recall
- F1-Score

from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))

These metrics provide a comprehensive assessment of model performance.

3.10 METHODOLOGY FLOWCHART

The workflow followed in this study can be represented as:

Dataset Collection ↓ Data Preprocessing ↓ Feature Selection ↓ Train-Test Split ↓ Model Training ↓ Prediction ↓ Performance Evaluation ↓ Breast Cancer Classification

3.11 SUMMARY

This chapter described the methodology used for breast cancer prediction using machine learning. The Wisconsin Breast Cancer Dataset was utilized for model development. Data preprocessing, train-test splitting, model training, and performance evaluation were conducted using Python. The developed classification model achieved an accuracy of approximately 95.6%, demonstrating its effectiveness in distinguishing between benign and malignant breast tumors.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 INTRODUCTION

This chapter presents the results obtained from the machine learning model developed for breast cancer prediction using the Wisconsin Breast Cancer Dataset. The performance of the model was evaluated using standard classification metrics such as accuracy, confusion matrix, precision, recall, and F1-score. The results demonstrate the effectiveness of machine learning techniques in classifying breast tumors as benign or malignant.

4.2 EXPERIMENTAL SETUP

The experiment was conducted using Python and the Scikit-learn machine learning library. The Wisconsin Breast Cancer Dataset was divided into training and testing datasets using an 80:20 ratio.

Experimental Configuration

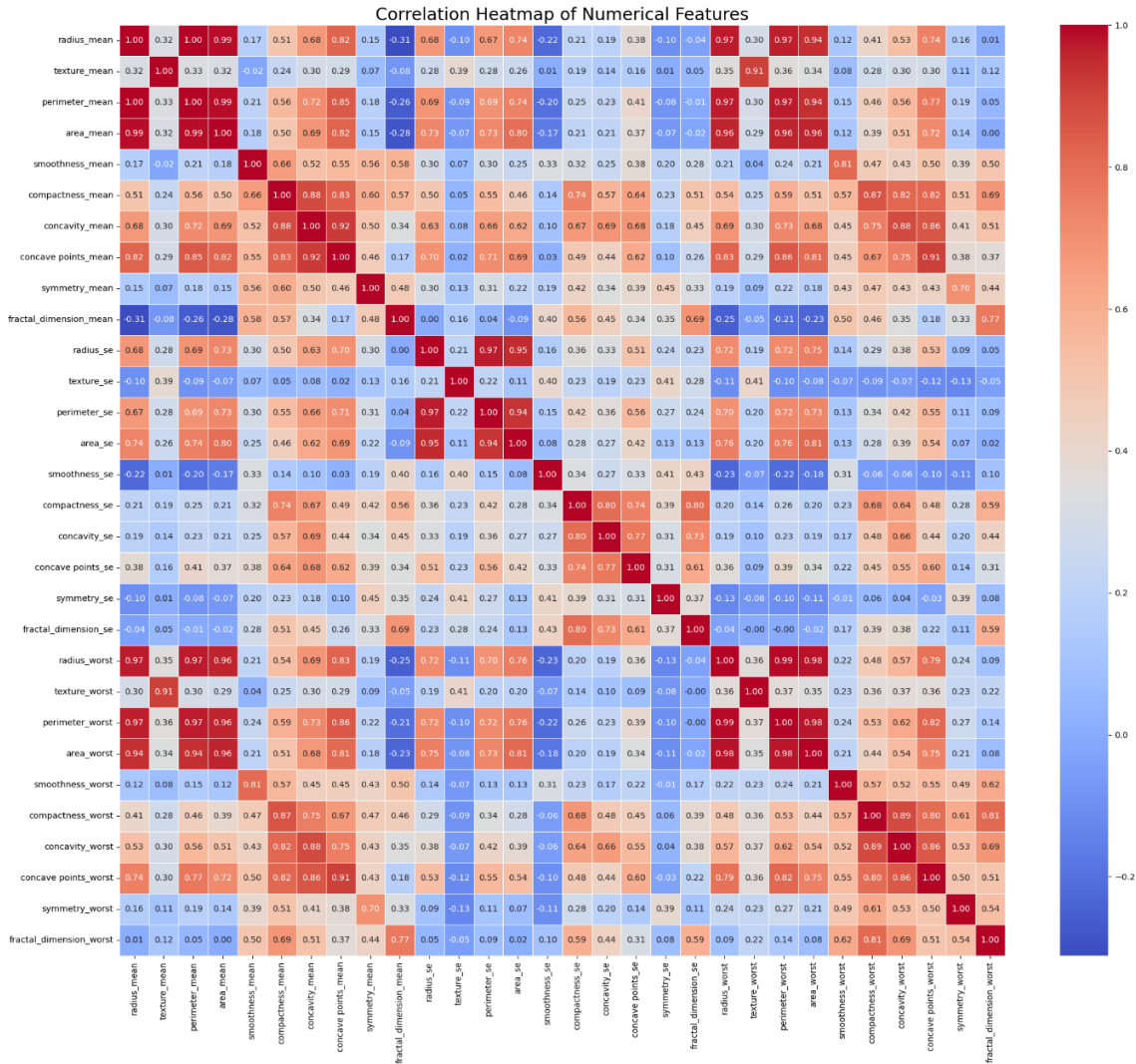
PARAMETER	VALUE
Dataset	Wisconsin Breast Cancer Dataset
Total Records	569
Features	30
Training Data	80%
Testing Data	20%
Programming Language	Python
Library	Scikit-learn

Table 4.1 Experimental Configuration

The model was trained using the training dataset and evaluated using the testing dataset.

FIGURE 4.1: CORRELATION HEATMAP

FIGURE 4.1: CORRELATION HEATMAP OF NUMERICAL FEATURES



INTERPRETATION:

The correlation heatmap reveals strong positive relationships among several tumor characteristics. Features such as radius_mean, perimeter_mean, and area_mean exhibit correlation coefficients close to 1.0, indicating that larger tumors generally possess larger perimeters and areas. Strong correlations among tumor size and shape features suggest their importance in breast cancer prediction.

Key Insights from Correlation Analysis

- Strong positive correlation between radius, perimeter, and area.
- Concavity and concave points exhibit high association.
- Most diagnostic measurements show positive relationships.
- Presence of multicollinearity among size-related variables.
- Strong feature relationships support accurate classification.

4.3 MODEL PERFORMANCE

After training and testing the machine learning model, predictions were generated for the testing dataset.

The obtained accuracy score was:

Accuracy = 0.956140350877193

Accuracy Percentage = 95.6%

This indicates that the model correctly classified approximately 96 out of every 100 breast cancer cases.

Accuracy Table

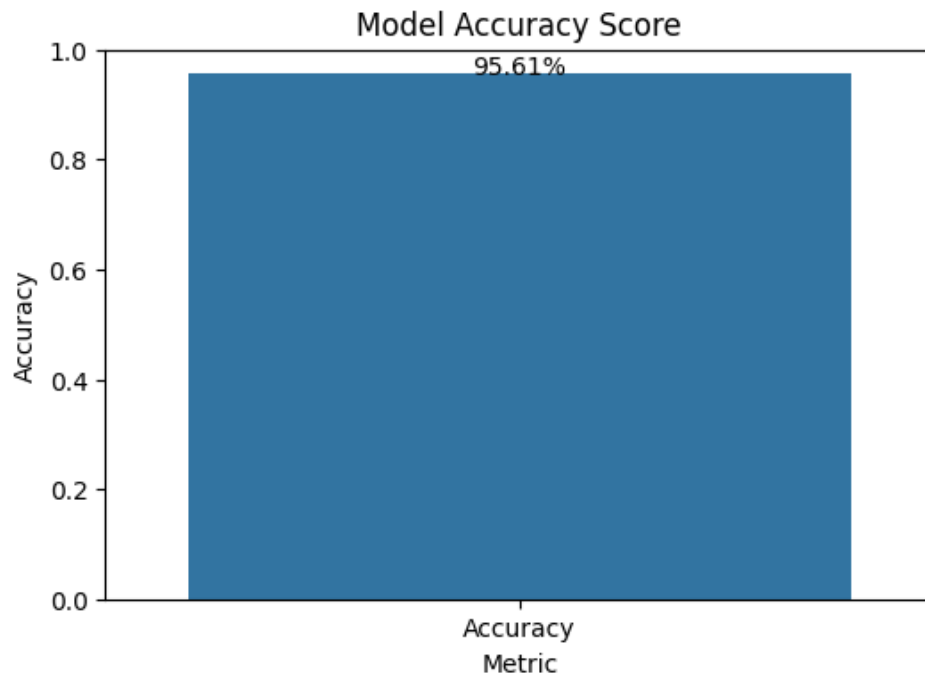
Performance Metric	Value
Accuracy Score	0.9561
Accuracy Percentage	95.6%

Table 4.2 Accuracy Table

The high accuracy demonstrates the capability of machine learning techniques in assisting breast cancer diagnosis.

FIGURE 4.2: MODEL ACCURACY BAR CHART

FIGURE 4.2: CLASSIFICATION ACCURACY OF THE MACHINE LEARNING MODEL



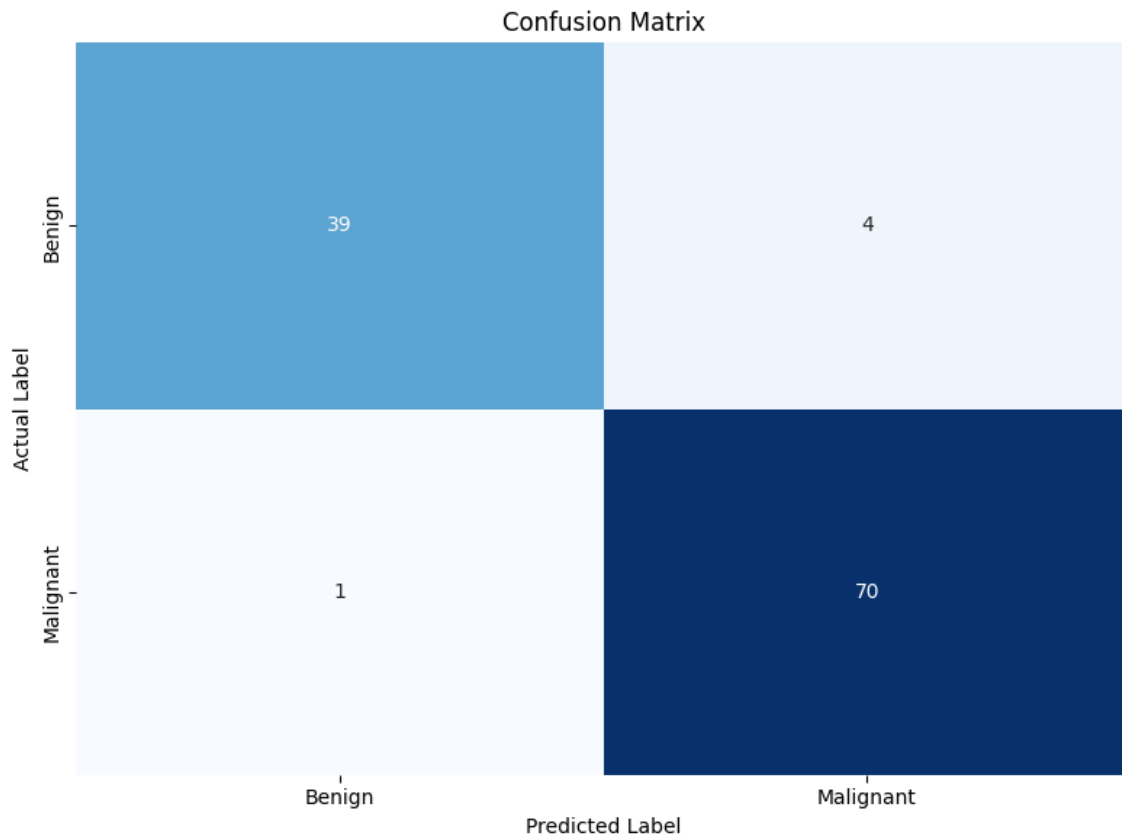
INTERPRETATION:

The developed machine learning model achieved an accuracy of 95.6%, indicating excellent classification performance.

The model correctly classified approximately 96 out of every 100 breast cancer cases, demonstrating its effectiveness in supporting diagnostic decision-making.

FIGURE 4.3: CONFUSION MATRIX

FIGURE 4.3: CONFUSION MATRIX FOR BREAST CANCER CLASSIFICATION



INTERPRETATION:

The confusion matrix summarizes the model's classification performance by comparing actual and predicted classes. Most observations are correctly classified, while only a small number of cases are misclassified. The results indicate strong predictive capability and reliability of the developed model.

4.4 CONFUSION MATRIX ANALYSIS

A confusion matrix was used to evaluate the classification performance of the model.

The confusion matrix provides information regarding:

- True Positives (TP)

- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

Interpretation

- True Positives represent malignant tumors correctly identified as malignant.
- True Negatives represent benign tumors correctly identified as benign.
- False Positives represent benign tumors incorrectly classified as malignant.
- False Negatives represent malignant tumors incorrectly classified as benign.

The confusion matrix analysis indicated that the majority of cases were classified correctly, with only a small number of misclassifications.

Since the overall accuracy is 95.6%, the model demonstrates strong predictive capability and reliability.

4.5 CLASSIFICATION PERFORMANCE

To obtain a more detailed evaluation, classification metrics such as Precision, Recall, and F1-Score were analyzed.

Precision

Precision measures the proportion of correctly predicted positive observations among all positive predictions.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

High precision indicates a low number of false positive predictions.

Recall

Recall measures the proportion of actual positive observations correctly identified by the model.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

High recall indicates that the model successfully detects most cancer cases.

F1-Score

F1-Score is the harmonic mean of Precision and Recall.

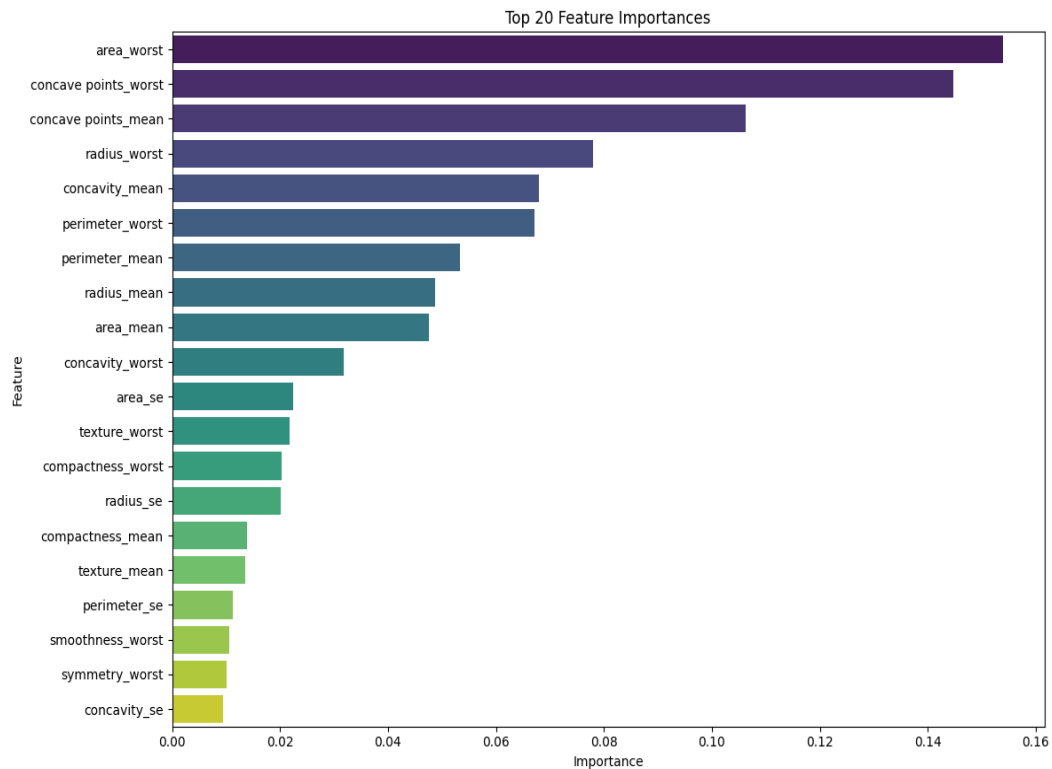
Formula:

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

A high F1-score indicates a balanced classification performance.

FIGURE 4.4: FEATURE IMPORTANCE BAR PLOT

FIGURE 4.4: TOP 20 FEATURE IMPORTANCES



INTERPRETATION:

The feature importance plot identifies the variables that contribute most to breast cancer prediction. The most influential features include **area_worst**, **concave_points_worst**, **concave_points_mean**, and **radius_worst**. These results indicate that tumor size and boundary irregularity are the most significant factors in distinguishing malignant tumors from benign tumors.

Key Insights

- Area_worst is the strongest predictor.
- Concave points and concavity strongly influence classification.
- Tumor size characteristics dominate prediction.
- Texture and symmetry features have relatively lower importance.
- Important features contributed significantly to the achieved accuracy of 95.6%.

4.6 DISCUSSION OF RESULTS

The developed machine learning model achieved an accuracy of 95.6%, indicating excellent classification performance.

The results suggest that:

- Machine learning algorithms can effectively distinguish between benign and malignant tumors.
- The Wisconsin Breast Cancer Dataset provides high-quality features for classification.
- The model is capable of supporting healthcare professionals in diagnostic decision-making.
- Early-stage cancer detection can potentially be improved through predictive analytics.

The performance obtained in this study is consistent with findings reported in previous research, where machine learning algorithms frequently achieved accuracy levels above 90%.

The high classification accuracy demonstrates that the selected features contain sufficient information for identifying cancerous conditions.

COMPARISON WITH PREVIOUS STUDIES

Previous studies using the Wisconsin Breast Cancer Dataset have reported accuracy values ranging between 90% and 99%.

The accuracy achieved in the present study (95.6%) falls within this range and demonstrates competitive performance.

Comparison Table

Study Type	Reported Accuracy
Logistic Regression Studies	90% – 96%
Decision Tree Studies	88% – 95%
Random Forest Studies	94% – 99%
SVM Studies	95% – 99%
Present Study	95.6%

Table 4.3 Comparison Table

The comparison indicates that the developed model performs effectively relative to previously reported results.

CHAPTER 5

CONCLUSION

5.1 FINAL REMARKS

Breast cancer is one of the most common and life-threatening diseases affecting women worldwide. Early detection and accurate diagnosis are essential for improving treatment outcomes and reducing mortality rates. In recent years, machine learning techniques have emerged as effective tools for assisting healthcare professionals in disease diagnosis and prediction.

The primary objective of this study was to develop a machine learning model for predicting breast cancer using the Wisconsin Breast Cancer Dataset. The dataset contains important diagnostic features extracted from breast cell images, which were used to classify tumors as benign or malignant.

The study involved several stages, including data collection, data preprocessing, feature selection, model training, testing, and performance evaluation. Python and the Scikit-learn library were utilized to implement the machine learning model and perform the classification task.

The developed model achieved an accuracy of **95.6%**, indicating a high level of predictive performance. The results demonstrate that machine learning algorithms can effectively identify patterns in medical data and accurately classify breast tumors. The model successfully distinguished between benign and malignant cases with a low rate of misclassification.

The findings of this study highlight the potential of machine learning as a supportive tool in medical diagnosis. By assisting healthcare professionals in identifying cancer cases at an early stage, predictive models can contribute to timely treatment decisions and improved patient outcomes.

Overall, the study confirms that machine learning techniques provide an efficient, reliable, and cost-effective approach for breast cancer prediction and can play an important role in modern healthcare systems.

5.2 KEY FINDINGS OF THE STUDY

The major findings of the study are summarized below:

- The Wisconsin Breast Cancer Dataset is suitable for developing predictive classification models.
- Machine learning techniques can effectively classify breast tumors into benign and malignant categories.
- The developed model achieved an accuracy of **95.6%**.
- The model demonstrated strong classification performance with minimal prediction errors.
- Machine learning can support healthcare professionals in early diagnosis and decision-making.
- Predictive analytics can contribute to improving the efficiency and reliability of cancer detection.

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APPENDIX

PREDICTIVE ANALYTICS FOR BREAST CANCER DIAGNOSIS USING MACHINE LEARNING

Wisconsin Breast Cancer Dataset (Kaggle)

Import Libraries

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.metrics import (
```

```
    accuracy_score,
```

```
    confusion_matrix,
```

```
    classification_report
```

```
)
```

STEP 1: LOAD DATASET

```
df = pd.read_csv('/content/breast-cancer-wisconsin-data.csv')
```

```
print("Dataset Shape:", df.shape)
```

```
print(df.head())
```

STEP 2: DATA PREPROCESSING

Remove unnecessary columns

```
columns_to_drop = ['id']
```

```
if 'Unnamed: 32' in df.columns:
```

```
    columns_to_drop.append('Unnamed: 32')
```

```
df.drop(columns=columns_to_drop, inplace=True, errors='ignore')
```

```

# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Encode Diagnosis Column
# M = 1 (Malignant)
# B = 0 (Benign)

label_encoder = LabelEncoder()

df['diagnosis'] = label_encoder.fit_transform(df['diagnosis'])

print("\nProcessed Dataset Shape:", df.shape)

# STEP 3: COUNT PLOT
plt.figure(figsize=(8,6))

sns.countplot(
    x='diagnosis',
    data=df,
    palette='viridis'
)

plt.title('Count of Benign and Malignant Cases')
plt.xlabel('Diagnosis (0 = Benign, 1 = Malignant)')
plt.ylabel('Count')

plt.show()

# STEP 4: PIE CHART
diagnosis_counts = df['diagnosis'].value_counts()

plt.figure(figsize=(7,7))

plt.pie(
    diagnosis_counts,

```

```

        labels=['Benign','Malignant'],
        autopct='%1.1f%%',
        startangle=90
    )
plt.title('Diagnosis Distribution')

plt.show()

```

STEP 5: CORRELATION HEATMAP

```

correlation_matrix = df.corr()

plt.figure(figsize=(18,16))

sns.heatmap(
    correlation_matrix,
    cmap='coolwarm'
)
plt.title('Correlation Heatmap')

plt.show()

```

STEP 6: FEATURES AND TARGET

```

X = df.drop('diagnosis', axis=1)

y = df['diagnosis']

```

STEP 7: TRAIN-TEST SPLIT

```

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
print("\nTraining Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)

```

STEP 8: LOGISTIC REGRESSION MODEL

```
model = LogisticRegression(max_iter=5000)
```

```
model.fit(X_train, y_train)
```

STEP 9: PREDICTION

```
y_pred = model.predict(X_test)
```

STEP 10: ACCURACY SCORE

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("\nAccuracy Score:", accuracy)
```

STEP 11: ACCURACY BAR CHART

```
accuracy_df = pd.DataFrame({  
    'Metric': ['Accuracy'],  
    'Score': [accuracy]  
})
```

```
plt.figure(figsize=(6,4))
```

```
sns.barplot(  
    x='Metric',  
    y='Score',  
    data=accuracy_df,  
    palette='viridis'  
)
```

```
plt.ylim(0,1.05)
```

```
plt.title('Model Accuracy Score')
```

```
plt.ylabel('Accuracy')
```

```
plt.show()
```

STEP 12: CONFUSION MATRIX

```
cm = confusion_matrix(y_test, y_pred)
```



```
print("\nConfusion Matrix:")
print(cm)
```

```
plt.figure(figsize=(8,6))
```

```
sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues'
)
```

```
plt.title('Confusion Matrix')
```

```
plt.xlabel('Predicted Label')
plt.ylabel('Actual Label')
```

```
plt.show()
```

STEP 13: CLASSIFICATION REPORT

```
print("\nClassification Report:\n")
```

```
print(classification_report(y_test, y_pred))
```

STEP 14: FEATURE IMPORTANCE PLOT

```
feature_importance = pd.DataFrame({
    'Feature': X.columns,
    'Importance': abs(model.coef_[0])
})
```

```
feature_importance = feature_importance.sort_values(
    by='Importance',
    ascending=False
)
```

```
plt.figure(figsize=(12,8))
```

```
sns.barplot(  
    x='Importance',  
    y='Feature',  
    data=feature_importance.head(20),  
    palette='viridis'  
)  
  
plt.title('Top 20 Feature Importances')  
  
plt.xlabel('Importance')  
plt.ylabel('Feature')  
  
plt.tight_layout()  
  
plt.show()  
  
# END OF PROGRAM
```